

Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customers segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary

- Rows : 3,900
- Columns : 18
- **Key Features**
 - Customer Demographics (Age, Gender, Location, Subscription Status).
 - Purchase Details (item *Purchased* , *Category* , *Purchase Amount* , *Season* , *Size* , *Color*)
 - Shopping Behavior (*Discount Applied* , *Promo Code Used* , *Previous Purchases*, *Frequency of Purchases*, *Review Rating*, *Shipping Type*)
 - Missing Data : 37 values in Review Rating Column

Exploratory Data Analysis Using Python

We began with data preparation and cleaning in Python:

- **Data Loading** : Imported the dataset using pandas.
- **Initial Exploration** : Used `df.info()` to check structure and `.describe()` for summary statistics.

df.describe(include='all')																							
	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases					
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900					
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7					
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	Paypal	Every 3 Months					
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584					
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN					
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN					
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN					
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN					
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN					
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN					
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN					

- **Missing Data Handling** : Checked for null values and imputed missing values in the `Review Rating` Column using the median rating of each product category.
- **Column Standardization** : Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering** :
 - Created `age_group` column by binning customers ages
 - Created `purchase_frequency_days` column from purchase data.
 - **Data Consistency Check** : Verified if `discount_applied` and `promo_code_used` were redundant, dropped `promo_code_used` .
 - **Database Integration** : Connected Python script to PostgreSQL and loaded the cleaned data into the database for SQL analysis.

Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** : Compared total revenue generated by male vs. female customers.

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